



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

November 2005

First Assessment

Mathematics

Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used
- All necessary working should be shown in every question

Total Marks – 60

- All Questions may be attempted
- Each Question is worth 15 marks, and should be handed up in a separate examination Booklet.

Examiner – A.M.Gainford

Question 1. (15 Marks) (Start a new booklet.)

Marks

- (a) The point $P(6, 9)$ divides the interval AB externally in the ratio 3:2. Find the coordinates of the point B given that A is $(1, 4)$. **2**
- (b) The derivative function of a curve is given by $\frac{dy}{dx} = 2x - 3$. If the curve passes through the point $(-1, -3)$, find its equation. **2**
- (c) Find $\frac{dy}{dx}$ given: **6**
- (i) $y = \sqrt{3x^2 - 4}$
- (ii) $y = x^3(2 - x)^4$
- (iii) $y = \frac{(x-1)^2}{2x+1}$
- (d) Solve the inequalities **5**
- (i) $\frac{1}{x} > \frac{1}{5}$
- (ii) $\frac{x}{x-1} \leq 2$

Question 2. (15 Marks) (Start a new booklet.)

Marks

- (a) Find a general solution of the equation

2

$$\sin \theta \cos \theta = \frac{1}{2}.$$

- (b) Given the polynomial $P(x) = x^3 + 2x^2 - 11x - 12$:

4

(i) Use the factor theorem to determine the zeroes of the polynomial.

(ii) Express $P(x)$ as a product of linear factors.

- (c) Prove the identity

2

$$\frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$$

- (d) Find the acute angle between the lines $x - 2y + 1 = 0$ and $3x + 4y - 3 = 0$, correct to the nearest minute. **3**

- (e) Fred wishes to purchase an annuity which will pay him \$28 000 at the end of each year, for a period of 20 years. The ruling interest rate is $9\frac{1}{2}\%$ per annum compound. How much will Fred have to pay for the annuity? **4**

Question 3. (15 Marks) (Start a new booklet.)

Marks

- (a) Solve the equation $2x^3 + 3x^2 - 3x - 2 = 0$ given that the roots are in geometric progression. **4**
- (b) From a point A on the ground the angle of elevation of the top (T) of a tower is 35° . **5**
- (i) Find the angle of elevation of the top of the tower from a point B on the same level as A , but twice as far from the base (X) of the tower.
- (ii) If A is due south of the tower and B is 130 metres due west of A , sketch a diagram to represent this situation.
- (iii) Find the height of the tower to the nearest metre.
- (c) (i) Show that the point $P(4t, 2t^2)$ lies on a parabola for all values of t , and state its focus and directrix. **4**
- (ii) Find the chord of contact to the parabola from the point $X(2, -2)$.
- (iii) Show that this chord passes through the focus of the parabola.
- (d) Use the t -method (where $t = \tan \frac{\theta}{2}$) to solve $\sin \theta + 3 \cos \theta = 2$ where $-180^\circ \leq \theta \leq 180^\circ$. Express your answer correct to the nearest minute. **2**

Question 4. (15 Marks) (Start a new booklet.)

Marks

(a) Consider the polynomial $P(x) = x^4 + x^3 - 8x^2 - ax - 9$: **4**

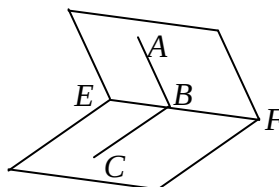
- (i) Find the value of the constant a if $P(x)$ is divisible by $Q(x) = x^2 - 9$.
- (ii) Show that for this value of a the only real zeroes of $P(x)$ are those of $Q(x)$.

(b) P and Q are points on the parabola with parametric equations $x = 2at$, $y = at^2$, where $t = p$ and $t = q$ respectively. **7**

- (i) State the coordinates of M , the midpoint of PQ , in terms of a , p and q .
- (ii) Show that if the chords PO and QO , where O is the origin, are perpendicular, then $pq = -4$.
- (iii) Hence show that the locus of M is a parabola, and state its vertex and focal length.
- (iv) Given that the equation of PQ is $a(p+q)x = 2a(y+apq)$ find the point of intersection of the tangents at P and Q .

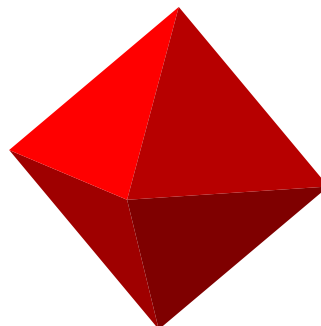
(c) It is given that the angle between two planes is the angle between two lines both perpendicular at the same point to the line of intersection of the planes.

$AB \perp EF$, $BC \perp EF$, and $\angle ABC$ is the angle between the planes.



Find the angle between two adjacent faces of a regular octahedron (in which all edges are of equal length).
Give your answer correct to the nearest minute.

4



End of paper



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Assessment Task #1

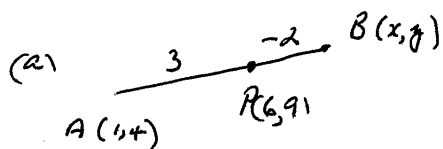
Mathematics

Extension 1

Sample Solutions

Question	Marker
1	PRB
2	DMH
3	CK
4	PSP

Question 1

(a) 

$$6 = \frac{3x-2}{3-2} \Rightarrow 3x=8 \therefore x=\frac{8}{3}$$

$$9 = \frac{3y-8}{3-2} \Rightarrow 3y=17 \therefore y=\frac{17}{3}$$

$$\therefore \underline{B \text{ is } \left(\frac{8}{3}, \frac{17}{3}\right).}$$

(b) N/A.

(c) (i) $y = \sqrt{3x^2-4}$
 $= (3x^2-4)^{\frac{1}{2}}$
 $y' = \frac{1}{2} (3x^2-4)^{-\frac{1}{2}} \times 6x$
 $\boxed{y' = \frac{3x}{\sqrt{3x^2-4}}}$

(ii) $y = x^3(2-x)^4$
 $y' = 3x^2(2-x)^4 + x^3 \cdot 4(2-x)^3 \cdot (-1)$
 $= x^2(2-x)^3 [3(2-x) - 4x]$
 $= x^2(2-x)^3 (6-3x-4x)$
 $\therefore \boxed{y' = x^2(2-x)^3 (6-7x)}$

(iii) $y = \frac{(x-1)^2}{2x+1}$
 $y' = \frac{(2x+1) \cdot 2(x-1) - (x-1)^2 \cdot 2}{(2x+1)^2}$
 $= \frac{2(x-1) [2x+1 - (x-1)]}{(2x+1)^2}$
 $\boxed{y' = \frac{2(x-1)(x+2)}{(2x+1)^2}}$

(iv) $\frac{x}{x-1} \leq 2$
 $\frac{x}{x-1} - 2 \leq 0$
 $\frac{x-2(x-1)}{x-1} \leq 0$
 $\frac{2-x}{x-1} \leq 0$ [x both sides by $(x-1)^2$]
 $(2-x)(x-1) \leq 0$

$$\therefore \boxed{x < 1, x \geq 2}$$

[NB $x \neq 1$]

* DO NOT WRITE AS

$$2 \leq x < 1 !!$$

(d) (i) $\frac{1}{x} > \frac{1}{5}$

$$\therefore \frac{5}{x} - 1 > 0$$

$$\frac{5-x}{x} > 0 \quad (\times \text{ both sides by } x^2)$$

$$x(5-x) > 0$$

$$\therefore \boxed{0 < x < 5}$$

Question 2

- (a) Find the general solution of the equation

$$\sin \theta \cos \theta = \frac{1}{2}.$$

Solution: $2 \sin \theta \cos \theta = 1,$
 $\sin 2\theta = 1,$
 $2\theta = n\pi + (-1)^n \frac{\pi}{2} \text{ where } n \in \mathbb{J},$
 $= \frac{\pi}{2} + 2n\pi,$
 $\theta = n\pi + \frac{\pi}{4}.$

- (b) Given the polynomial $P(x) = x^3 + 2x^2 - 11x - 12$:

- (i) Use the factor theorem to determine the zeroes of the polynomial.

Solution: Factors of 12 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$.

$$P(1) = 1 + 2 - 11 - 12 \neq 0,$$

$$P(-1) = -1 + 2 - 11 - 12 \neq 0.$$

$$\begin{array}{r} x^2 + x - 12 \\ x+1 \overline{) x^3 + 2x^2 - 11x - 12} \\ \underline{-x^3 - x^2} \\ x^2 - 11x \\ \underline{-x^2 - x} \\ -12x - 12 \\ \underline{12x + 12} \\ 0 \end{array}$$

$$\text{or } -1 \left| \begin{array}{cccc} 1 & 2 & -11 & -12 \\ & -1 & -1 & 12 \\ 1 & 1 & -12 & 0 \end{array} \right|$$

$$\text{Now } x^2 + x - 12 = (x+4)(x-3)$$

$\therefore$ zeroes are $-1, -4, 3$.

- (ii) Express $P(x)$ as a product of linear factors.

Solution: $P(x) = (x+1)(x+4)(x-3)$

- (c) Prove the identity

$$\frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$$

Solution: L.H.S. = $\frac{\sin^2 \theta + 1 - 2 \cos \theta + \cos^2 \theta}{\sin \theta(1 - \cos \theta)}$,
= $\frac{2(1 - \cos \theta)}{\sin \theta(1 - \cos \theta)}$,
= $\frac{2}{\sin \theta}$,
= $2 \operatorname{cosec} \theta$,
= R.H.S.

- (d) Find the acute angle between the lines $x - 2y + 1 = 0$ and $3x + 4y - 3 = 0$, correct to the nearest minute.

Solution: $m_1 = \frac{1}{2}$, $m_2 = -\frac{3}{4}$.
 $\tan \alpha = \left| \frac{\frac{1}{2} + \frac{3}{4}}{1 - \frac{1}{2} \times \frac{3}{4}} \right|$,
= $\left| \frac{4+6}{8-3} \right|$,
= 2.
 $\therefore$ Acute angle is $63^\circ 26'$ (nearest minute).

- (e) Fred wishes to purchase an annuity which will pay him \$28 000 at the end of each year, for a period of 20 years. The ruling interest rate is $9\frac{1}{2}\%$ per annum compound. How much will Fred have to pay for the annuity?

Solution: Let purchase price be \$P.
Value after 1 year = $P \times 1.095 - 28\,000$,
Value after 2 years = $(P \times 1.095 - 28\,000)1.095 - 28\,000$,
Value after 3 years = $((P \times 1.095 - 28\,000)1.095 - 28\,000)1.095 - 28\,000$,
= $P \times 1.095^3 - 28\,000(1 + 1.095 + 1.095^2)$,
= $P \times 1.095^3 - \frac{28\,000(1.095^3 - 1)}{0.095}$,
Value after 20 years = $P \times 1.095^{20} - \frac{28\,000(1.095^{20} - 1)}{0.095}$,
= 0.
 $\therefore$, $P \times 1.095^{20} = \frac{28\,000(1.095^{20} - 1)}{0.095}$,
So $P = \frac{28\,000(1.095^{20} - 1)}{1.095^{20} \times 0.095}$,
= 246 746.70 (2 d.p.)
 $\therefore$ Annuity will cost \$246 746.70.

Question 3

(a) Let roots be $\frac{\alpha}{\beta}, \alpha, \alpha\beta$

$$\frac{\alpha}{\beta} + \alpha + \alpha\beta = -\frac{3}{2} \quad \text{--- (1)}$$

$$\frac{\alpha^2}{\beta} + \alpha^2 + \alpha^2\beta = -\frac{3}{2} \quad \text{--- (2)}$$

$$\alpha^3 = 1 \quad \text{--- (3)}$$

From (3) $\alpha = 1$

$\Rightarrow$ (1) and (2) [same eqⁿ]

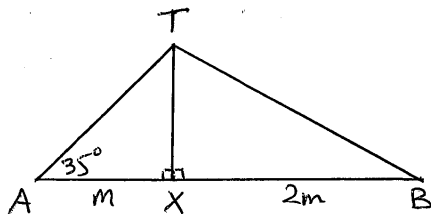
$$\text{i.e. } \frac{1}{\beta} + 1 + \beta = -\frac{3}{2}$$

$$2\beta^2 + 5\beta + 2 = 0$$

$$\therefore \beta = -2, -\frac{1}{2}$$

Roots are $1, -2, -\frac{1}{2}$

(b) (i)

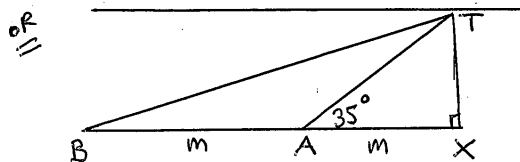


$$\tan 35^\circ = \frac{TX}{m} \quad \text{and} \quad \tan B = \frac{TX}{2m}$$

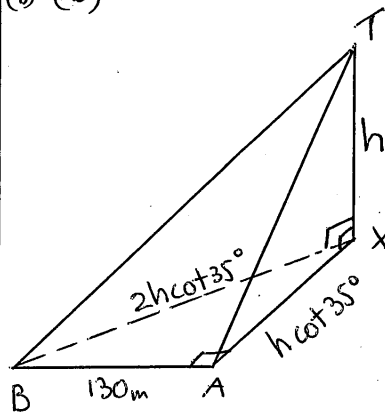
$$\Rightarrow m \tan 35^\circ = 2m \tan B$$

$$\frac{1}{2} \tan 35^\circ = \tan B$$

$$\text{i.e. } \hat{B} = 19^\circ 18'$$



(b) (ii)



(iii) In ΔABX by Pyth. Theorem

$$130^2 + (h \cot 35^\circ)^2 = (2h \cot 35^\circ)^2$$

$$h = \frac{130 \tan 35^\circ}{\sqrt{3}} \approx 53 \text{ m}$$

(c) (i) $x = 4t, y = 2t^2$

$$\Rightarrow t = \frac{x}{4} \therefore y = 2\left(\frac{x}{4}\right)^2$$

$$\text{i.e. } x^2 = 8y$$

Focus $(0, 2)$ DIR. $y = -2$

(ii) Eqⁿ of C of C is

$$xx_0 = 2a(y + y_0)$$

$$\therefore 2x = 4(y - 2)$$

$$x = 2y - 4$$

(iii) Focus $(0, 2)$

$$\Rightarrow 0 = 2(2) + 4$$

pt Satisfies eqⁿ of C of C

$\therefore$ lies on $x = 2y - 4$

(d)

$$\sin \theta + 3 \cos \theta = 2$$

$$\frac{2t}{1+t^2} + 3 \left[\frac{1-t^2}{1+t^2} \right] = 2$$

$$2t + 3 - 3t^2 = 2 + 2t^2$$

$$5t^2 - 2t - 1 = 0$$

$$t = \frac{1 \pm \sqrt{6}}{5}$$

$$\tan \frac{\theta}{2} = \frac{1 \pm \sqrt{6}}{5}$$

$$\frac{\theta}{2} = 34^\circ 36', -16^\circ 10'$$

$$\theta = 69^\circ 12', -32^\circ 20'$$

NB Clearly $\theta = 180^\circ$ is not a solution of the original equation

Question 4

$$\begin{aligned} \text{(a)} \quad \text{(i)} \quad P(x) &= x^4 + x^3 - 8x^2 - ax - 9 \\ &= (x^2 - 9)Q(x) \\ \therefore P(3) &= P(-3) = 0 \quad (\text{Factor Theorem}) \\ P(3) &= 3^4 + 3^3 - 8(3)^2 - 3a - 9 = 0 \Rightarrow a = 9 \\ P(-3) &= (-3)^4 + (-3)^3 - 8(-3)^2 + 3a - 9 = 0 \Rightarrow a = 9 \\ \therefore a &= 9 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(x) &= (x^2 - 9)(x^2 + Bx + 1) \\ P(1) &= -24 \\ \therefore -24 &= (-8)(2 + B) \Rightarrow B = 1 \\ \therefore P(x) &= (x^2 - 9)(x^2 + x + 1) \end{aligned}$$

For $P(x) = 0$ then either

1. $x^2 - 9 = 0 \Rightarrow x = \pm 3$ **OR**
2. $x^2 + x + 1 = 0 \Rightarrow \Delta = -3$

Clearly with $\Delta = -3$ then $x^2 + x + 1 = 0$ has no **REAL** solutions.
So the only real **zeroes** of $P(x)$ are those of $Q(x) = x^2 - 9$

$$\begin{aligned} \text{(b)} \quad &P(2ap, ap^2), Q(2aq, aq^2) \\ \text{(i)} \quad &M\left\{a(p+q), a\left(\frac{p^2+q^2}{2}\right)\right\} \\ \text{(ii)} \quad &OP \perp OQ \Rightarrow m_{OP} \times m_{OQ} = -1 \\ &m_{OP} = \frac{ap^2}{2ap} = \frac{p}{2} \\ &\therefore m_{OQ} = \frac{q}{2} \\ &\therefore \frac{p}{2} \times \frac{q}{2} = -1 \Rightarrow pq = -4 \end{aligned}$$

$$\begin{aligned}
\text{(iii)} \quad x &= a(p+q) \Rightarrow \frac{x}{a} = p+q \\
y &= a\left(\frac{p^2+q^2}{2}\right) \Rightarrow \frac{2y}{a} = p^2+q^2 \\
\left(\frac{x}{a}\right)^2 &= (p+q)^2 = p^2+q^2+2pq = \frac{2y}{a} - 8 \\
\therefore x^2 &= 2ya - 8a^2 = 2a(y-4a)
\end{aligned}$$

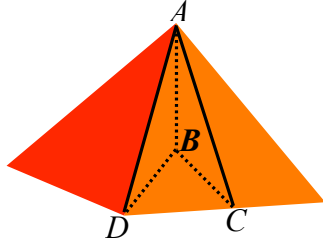
Clearly this is the equation of a parabola with vertex $(0, 4a)$ with a focal length of $\frac{1}{2}|a|$

(iv) Clearly PQ is the chord of contact from the point $T(x_0, y_0)$ where $xx_0 = 2a(y+y_0)$ and $T(x_0, y_0)$ is the point of intersection of the tangents from P and Q .

$$\therefore a(p+q)x = 2a(y+apq) \Rightarrow \begin{cases} x_0 = a(p+q) \\ y_0 = apq \end{cases}$$

So T is the point $\{a(p+q), apq\}$ or $\{a(p+q), -4a\}$ ($\because pq = -4$)

(c)



The angle we require is $\theta = 2 \times \angle ACB$.
Let the sides of the octahedron be $2a$ units.

$$\therefore DC = BC = a \Rightarrow DB = a\sqrt{2} \quad (\text{Pythagoras' Theorem})$$

$$\therefore AB = \sqrt{4a^2 - 2a^2} = a\sqrt{2} \quad (\text{Pythagoras' Theorem})$$

$$\therefore \tan(\angle ACB) = \frac{a\sqrt{2}}{a} = \sqrt{2}$$

$$\therefore \theta = 2 \tan^{-1} \sqrt{2} = 109^\circ 28'$$